

Number formatting in Excel changes how a value looks without changing the actual number stored in the cell. This is essential for ensuring that your data, whether it's for a **Salary statement** or an **Engineering project**, is readable and professional.

1. Common Number Formats

Excel provides several built-in categories to handle different types of data:

- **General:** The default format that displays numbers exactly as you type them.
- **Number:** Used for general display of numbers with a fixed number of decimal places and optional 1000-separator (e.g., **2,500.00**).
- **Currency & Accounting:** Adds a symbol like ₹ or \$. The "Accounting" format aligns the symbols and decimal points in a column, which is better for financial reports.
- **Date & Time:** Converts serial numbers into readable dates (e.g., **06-05-2026**).
- **Percentage:** Multiplies the cell value by 100 and displays it with a % symbol (e.g., **0.92** becomes **92%**).

2. Scientific and Special Formats

For specialized technical work, such as **Deep Learning** or **Data Science** engineering:

- **Scientific:** Used for very large or small numbers (e.g., **1.05E+06** for **1,048,576 rows**).
- **Fraction:** Displays numbers as actual fractions (e.g., **0.5** as **1/2**) instead of decimals.
- **Special:** Includes preset formats like Zip Codes or Phone Numbers.

3. Custom Number Formatting

If the standard options don't fit, you can create your own formatting codes.

- **Syntax:** A custom code can have up to four sections: [Positive]; [Negative]; [Zero]; [Text].
- **Placeholders:**
 - **0:** Forces a digit to be displayed (e.g., **0.00** always shows two decimals).
 - **#:** Only shows a digit if it is necessary.
 - **"Text":** You can add units by putting text in quotes (e.g., **0 "kg"**).

4. Tools for Fast Formatting

- **The Ribbon:** Use the dropdown menu and quick buttons (%, \$, .) in the **Number** group on the **Home** tab.
- **Increase/Decrease Decimal:** Quickly adjust the precision of your results.

- **Format Painter:** Copy the number formatting from one cell and "paint" it onto others.
- **Shortcuts:**
 - **Ctrl + Shift + !:** Apply Number format.
 - **Ctrl + Shift + \$:** Apply Currency format.
 - **Ctrl + Shift + %:** Apply Percentage format.

5. Data Integrity and Formatting

- **Formatting vs. Value:** Remember that formatting only changes the *display*. A cell formatted to show **0** decimals might show **10**, but Excel will still use **9.8** in any calculations unless you use the **ROUND** function.
- **Data Validation:** You can use validation to ensure users enter numbers in the correct format initially.
- **Conditional Formatting:** You can change a number's color based on its value (e.g., red for negative balances in a **Salary statement**).